

Table 12.2 Example 12.3 estimation errors (averaged over 100 Monte Carlo simulations) of the unconstrained and constrained Kalman and H_∞ filters with off-nominal noise statistics. The H_∞ filters perform better for position estimation, and the Kalman filters perform better for velocity estimation. Position errors are in units of meters, and velocity errors are in units of meters/second.

	Kalman		H_∞	
	Pos.	Vel.	Pos.	Vel.
Unconstrained	60.8	19.2	45.9	20.6
Constrained	56.2	17.6	39.1	19.1

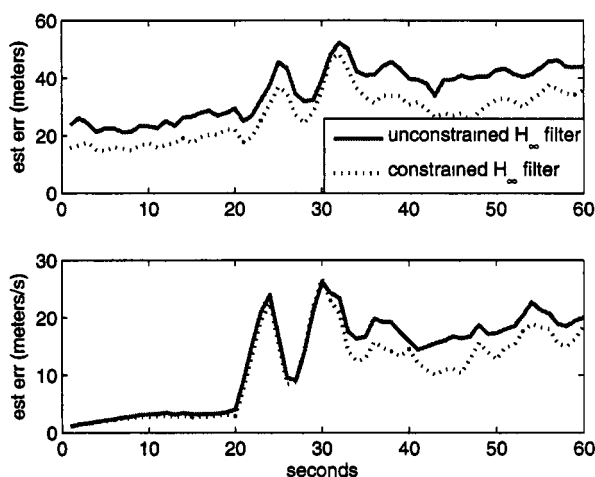


Figure 12.4 Example 12.3 unconstrained and constrained H_∞ filter estimation-error magnitudes. The plots show the average estimation-error magnitudes of 100 Monte Carlo simulations when the noise statistics are nominal.

12.4 SUMMARY

In this chapter we briefly introduced some advanced topics in the area of H_∞ filtering. We discussed an approach for minimizing a combination of the Kalman and H_∞ filter performance indices. This provides a way to balance the excessive optimism of the Kalman filter with the excessive pessimism of the H_∞ filter. We also looked at the robust mixed Kalman/ H_∞ estimation problem, where we took system-model uncertainties into account. This is an important problem because (in practice) the system model is never perfectly known. Finally we discussed constrained H_∞ filtering, in which equality (or inequality) constraints are enforced on the state estimate. This can improve filter performance in cases in which we know that the state must satisfy certain constraints.

There is still a lot of room for additional work and development in H_∞ filtering. For example, reduced-order H_∞ filtering tries to obtain good minimax estimation performance with a filter whose order is less than that of the underlying system.

Reduced-order Kalman filtering was discussed in Section 10.3, and reduced-order H_∞ filtering is considered in [Bet94, Gri97, Xu02]. The approach taken in [Ko06] for constrained Kalman filtering may be applicable to constrained H_∞ filtering and may give better results than the method discussed in this chapter. The use of Krein space approaches for solving various H_∞ filtering problems is promising [Has96a, Has96b]. H_∞ smoothing is discussed in [Gri91a, The94b, Has99, Zha05a], and robust H_∞ smoothing is discussed in [The94a]. An information form for the H_∞ filter (analogous to the Kalman information filter discussed in Section 6.2) is presented in [Zha05b]. Approaches to dealing with delayed measurements and synchronization errors have been extensively explored for Kalman filters (see Section 10.5), but are notably absent in the H_∞ filter literature. There has been a lot of work on nonlinear Kalman filtering (see Chapters 13–15), but not nearly as much on nonlinear H_∞ filtering.

PROBLEMS

Written exercises

12.1 Consider the system described in Example 12.1 with $Q = R = 1$.

- Find the steady-state *a priori* estimation-error variance P as a function of the estimator gain K .
- Find $\|G_{\tilde{x}e}\|_\infty^2$, the square of the infinity-norm of the transfer function from the noise w and v to the *a priori* state estimation error $\tilde{x}$, as a function of the estimator gain K .
- Find the estimator gain K that minimizes $(P + \|G_{\tilde{x}e}\|_\infty^2)$.

12.2 Verify that if $\theta = 0$, the Riccati equation associated with the mixed Kalman/ H_∞ filter in Equation (12.6) reduces to the Riccati equation associated with the Kalman filter.

12.3 Suppose that the hybrid filter gain of Equation (12.13) is used for the system of Example 12.1 with $\theta = 1/2$. For what values of d will the hybrid filter be stable?

12.4 Suppose that the robust filter of Section 12.2 is used for a system with n states and r measurements. What are the dimensions of M_1 , M_2 , Γ , and N ?

12.5 Suppose that a system matrix is given as

$$F = \begin{bmatrix} 0.4 \pm 0.2 & 0.4 \\ -0.4 & 1 \end{bmatrix}$$

(Note that this is the system matrix of Example 4.1 in case the effect of overcrowding on the predator population is uncertain.) Give an M_1 and N matrix that satisfy Equation (12.15) for this uncertainty.

12.6 Consider an uncertain system with $F = -1$, $H = 1$, $Q = R = 1$, $M_1 = 1/5$, $M_2 = 0$, and $N = 1$. Suppose that $\epsilon = 0$ is used to design a robust mixed Kalman/ H_∞ filter.

- For what values of α will the steady-state value of $\tilde{P}$ in Equation (12.22) be real and positive?

- b) For what values of α will the steady-state value of $\tilde{P}$ satisfy the second condition of Equation (12.24)?

12.7 Consider a constrained H_∞ state estimation problem with

$$\begin{aligned} F &= \begin{bmatrix} 1 & 1/2 \\ 1/2 & 0 \end{bmatrix} \\ G = H &= \begin{bmatrix} G_1 & 0 \end{bmatrix} \\ D &= \begin{bmatrix} 1 & 1 \end{bmatrix} \\ Q &= \begin{bmatrix} Q_1 & 0 \\ 0 & Q_2 \end{bmatrix} \end{aligned}$$

Find the steady-state constrained Riccati solution for P from Equation (12.50). For what values of G_1 will the condition of Equation (12.51) be satisfied?

Computer exercises

12.8 Consider a two-state Newtonian system as discussed in Example 9.1 with $T = 1$, $a = 1$, and $R = 1$.

- What is the steady-state Kalman gain?
- What is the maximum θ for which the H_∞ estimator exists? Answer to the nearest 0.01. What is the H_∞ gain for this value of θ ?
- What is the H_∞ gain when $\theta = 0.5$? Plot the maximum estimator eigenvalue magnitude as a function of d for the hybrid filter of Equation (12.13) when $\theta = 0.5$.

12.9 Implement the time-varying Riccati equations for the robust mixed Kalman/ H_∞ filter for $F = 1/2$, $H = Q = R = 1$, $M_1 = 1/4$, $M_2 = 0$, $N = 1$, $\epsilon = 0$, $\theta = 1/10$, and $S_1 = S_2 = 1$.

- At what time do the conditions of Equation (12.24) fail to be satisfied when $\alpha = 2$? Repeat for $\alpha = 3, 4, 5$, and 6 .
- What is the steady-state theoretical bound on the estimation error when $\alpha = 10$? Repeat for $\alpha = 20, 30$, and 40 .

12.10 Consider a constrained H_∞ state estimation problem with

$$\begin{aligned} F &= \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \\ H &= \begin{bmatrix} 1 & 0 \end{bmatrix} \\ G &= \begin{bmatrix} G_1 & 0 \end{bmatrix} \\ D &= \begin{bmatrix} 1 & 1 \end{bmatrix} \\ Q &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \end{aligned}$$

Implement the Σ_k and P_k expressions from Equation (12.50).

- What is the largest value of G_1 for which P_k reaches a positive definite steady-state solution that satisfies the condition given in Equation (12.51)? Answer to the nearest 0.01. What is the resulting steady-state value of P ?

- b) Set G_1 equal to 1% of the maximum G_1 that you found in part (a). What is the new steady-state value of P ? Give an intuitive explanation for why P gets smaller when G_1 gets smaller.

PART IV

NONLINEAR FILTERS
